

Lesson 07: Terraform Outputs

Learning Objectives

- Understand what output values are and why they are useful
- Define output blocks to expose resource attributes
- Use `terraform output` to retrieve values
- Mark outputs as sensitive
- Use outputs for cross-module references

1. What are Output Values?

Output values let you **export** information about your infrastructure after `terraform apply`. They are displayed in the CLI and queryable via `terraform output`.

```
output "bucket_name" {
  value = aws_s3_bucket.main.id
}
```

After apply, you see:

```
Outputs:

bucket_name = "my-bucket"
```

Why use outputs?

Use Case	Example
Quick access to ARNs and IDs	Get bucket ARN without browsing the Web UI
Pass data between modules	Feed module A's output into module B
Script automation	<code>terraform output -json</code> for programmatic access
Documentation	Show key resource identifiers after provisioning

2. Output Block Anatomy

```
output "output_name" {
  description = "Human-readable explanation"
  value       = <expression>
  sensitive   = true      # optional, hides from CLI
```

```
depends_on = []      # optional, explicit dependency
}
```

Argument	Description	Required
<code>value</code>	The expression to evaluate and return	Yes
<code>description</code>	Human-readable explanation	No
<code>sensitive</code>	If <code>true</code> , hides value in CLI (still accessible)	No (default <code>false</code>)
<code>depends_on</code>	Explicit dependency on other resources	No

3. Basic Outputs

```
# outputs.tf

output "s3_bucket_arn" {
  description = "ARN of the S3 bucket"
  value       = aws_s3_bucket.main.arn
}

output "s3_bucket_name" {
  description = "Name of the S3 bucket"
  value       = aws_s3_bucket.main.id
}

output "dynamodb_table_arn" {
  description = "ARN of the DynamoDB table"
  value       = aws_dynamodb_table.main.arn
}

output "dynamodb_table_name" {
  description = "Name of the DynamoDB table"
  value       = aws_dynamodb_table.main.name
}

output "sqs_queue_arn" {
  description = "ARN of the SQS queue"
  value       = aws_sqs_queue.main.arn
}

output "sqs_queue_url" {
  description = "URL of the SQS queue"
  value       = aws_sqs_queue.main.url
}
```

After `terraform apply`, you see:

Outputs:

```
dynamodb_table_arn    = "arn:aws:dynamodb:us-east-1:000000000000:table/AppTable"
dynamodb_table_name   = "AppTable"
s3_bucket_arn         = "arn:aws:s3:::my-bucket"
s3_bucket_name        = "my-bucket"
sqs_queue_arn         = "arn:aws:sqs:us-east-1:000000000000:my-queue"
sqs_queue_url         = "http://sqs.us-east-1:4566/localhost.localstack.cloud:4566/000000000000/my-queue"
```

4. Complex Output Expressions

Outputs can contain any Terraform expression, not just direct attribute references:

```
# Combined string
output "bucket_domain" {
  value = "${aws_s3_bucket.main.bucket_domain_name}"
}

# Conditional output
output "versioning_status" {
  value = aws_s3_bucket_versioning.main.versioning_configuration[0].status
}

# Map of all values
output "resource_identifiers" {
  value = {
    bucket_arn = aws_s3_bucket.main.arn
    table_arn  = aws_dynamodb_table.main.arn
    queue_arn  = aws_sqs_queue.main.arn
  }
}

# Formatted string
output "connection_string" {
  value = "endpoint=${aws_sqs_queue.main.url};region=us-east-1"
}
```

5. Sensitive Outputs

Use `sensitive = true` for values you don't want displayed in CLI output:

```
output "db_password" {
  description = "Database password"
  value       = var.db_password
  sensitive   = true
}
```

```
sensitive    = true
}
```

In the CLI:

Apply complete! Resources: 1 added.

Outputs:

db_password = (sensitive value)

You can still retrieve the value:

```
# Show all (sensitive values hidden)
terraform output

# Show sensitive values
terraform output db_password

# Show as JSON (includes sensitive)
terraform output -json
```

6. Using `terraform output` Command

```
# List all outputs
terraform output

# Get a specific output
terraform output bucket_domain

# Get in JSON format
terraform output -json

# Get raw value (no quotes)
terraform output -raw bucket_domain
```

Usage in Scripts

```
#!/bin/bash
BUCKET_NAME=$(terraform output -raw bucket_name)
QUEUE_URL=$(terraform output -raw sqs_queue_url)

aws --endpoint-url=http://localhost:4566 s3 ls s3://$BUCKET_NAME/
```

```
aws --endpoint-url=http://localhost:4566 sqs send-message \
  --queue-url $QUEUE_URL \
  --message-body "Automated message"
```

7. Outputs for Cross-Module References

When you split your config into modules, outputs become the **public API** of a module:

```
root/
├── main.tf
├── modules/
│   └── storage/
│       ├── main.tf
│       ├── outputs.tf    # <-- exposes bucket_arn, table_name
│       └── variables.tf
└── main.tf                # <-- uses module.storage.bucket_arn
```

Module outputs are accessed as `module.<module_name>.<output_name>`.

8. Complete Example

```
# outputs.tf
output "bucket_arn" {
  description = "ARN of the created S3 bucket"
  value       = aws_s3_bucket.main.arn
}

output "bucket_name" {
  description = "Name of the created S3 bucket"
  value       = aws_s3_bucket.main.id
}

output "table_arn" {
  description = "ARN of the DynamoDB table"
  value       = aws_dynamodb_table.main.arn
}

output "table_name" {
  description = "Name of the DynamoDB table"
  value       = aws_dynamodb_table.main.name
}

output "queue_arn" {
  description = "ARN of the SQS queue"
  value       = aws_sqs_queue.main.arn
}
```

```
output "queue_url" {
  description = "URL of the SQS queue"
  value       = aws_sqs_queue.main.url
}

output "all_resources" {
  description = "Map of all created resource identifiers"
  value = {
    bucket = aws_s3_bucket.main.arn
    table  = aws_dynamodb_table.main.arn
    queue  = aws_sqs_queue.main.arn
  }
}
```

9. Key Takeaways

- Outputs display resource attributes after **terraform apply**
- **terraform output -raw <name>** gets plain values for scripts
- **terraform output -json** returns all outputs as JSON
- **sensitive = true** hides values from CLI display but not from **terraform output -json**
- Outputs are the primary mechanism for passing data between modules
- Outputs can contain complex expressions: maps, lists, conditionals, string interpolation

example and exercise repo url

<https://github.com/eyu-se/terraform-training-material-with-projects>

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